SOFTWARE ENGINEERING LAB

LAB – 6

SEQUENCE DIAGRAM

Library Management System

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Course	B.Tech – Computer Science and Engineering
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Assignment	LAB-6: Draw the Sequence Diagram
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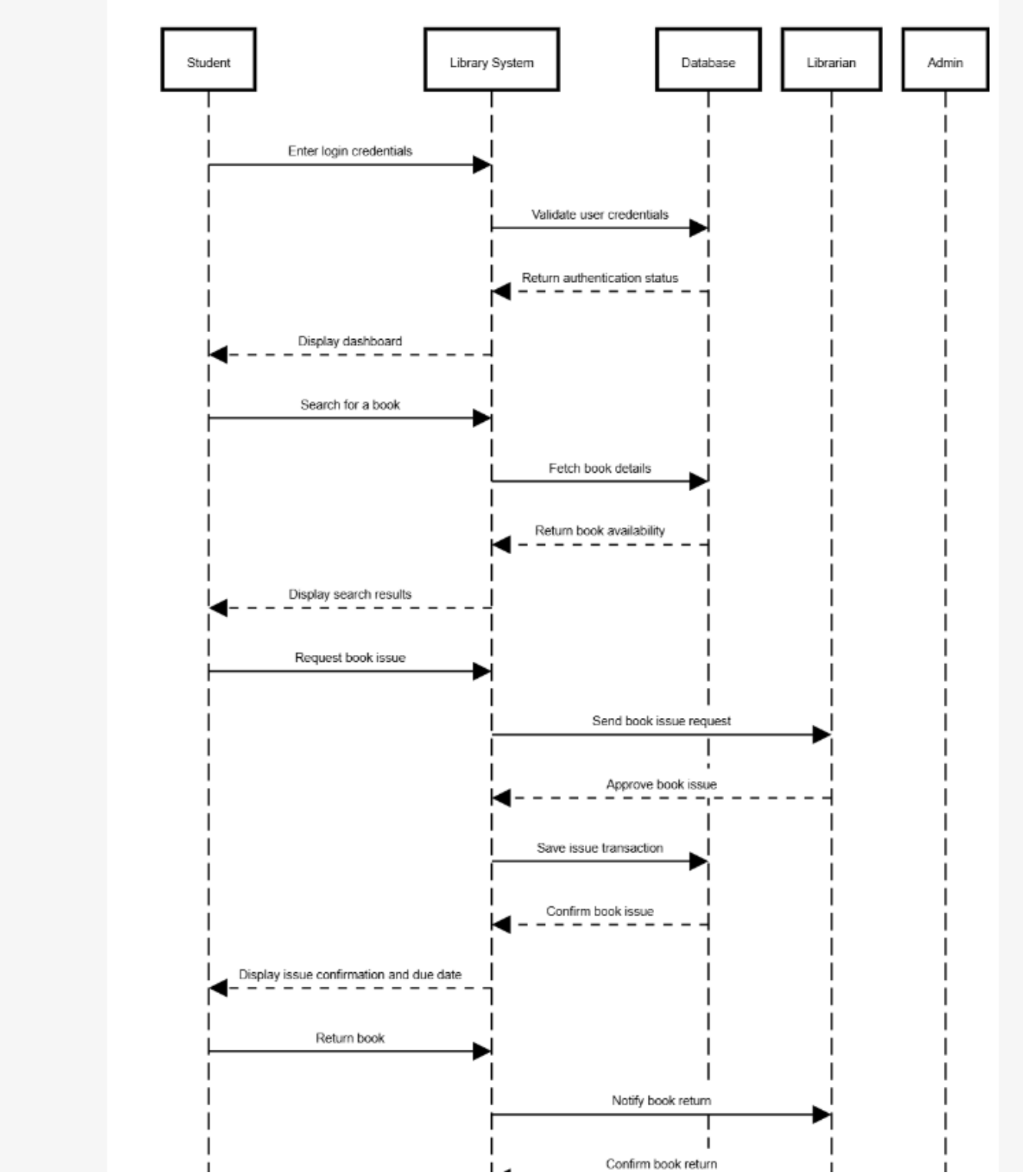
LAB–6: Draw the Sequence Diagram

Aim: To draw a UML sequence diagram for the Library Management System and represent the interactions between the Student, Library System, Database, Librarian, and Admin in chronological order.

Introduction: A sequence diagram is a UML interaction diagram that shows how different actors and system components communicate over time. It represents participants as lifelines and messages as horizontal arrows. The vertical direction indicates the order in which interactions occur.

Sequence Diagram

The following diagram is based on the reference image provided. It shows login validation, book searching, issuing a book, and the beginning of the return process.



Explanation of the Sequence

1. Login: The Student enters login credentials. The Library System sends them to the Database for validation. The Database returns the authentication status, and the system displays the dashboard.

2. Search for a book: The Student searches for a book. The Library System fetches book details from the Database, receives availability information, and displays the search results.

3. Issue a book: The Student requests a book issue. The Library System forwards the request to the Librarian. After approval, the system saves the issue transaction in the Database. The confirmation and due date are then displayed to the Student.

4. Return a book: The Student submits the book for return. The Library System notifies the Librarian, who confirms the return. The system then updates the return record in the Database. The supplied screenshot is cropped during this part of the sequence.

Conclusion: The sequence diagram illustrates the time-ordered communication among the Student, Library System, Database, Librarian, and Admin. It helps in understanding the workflow and responsibilities of each participant in the Library Management System.